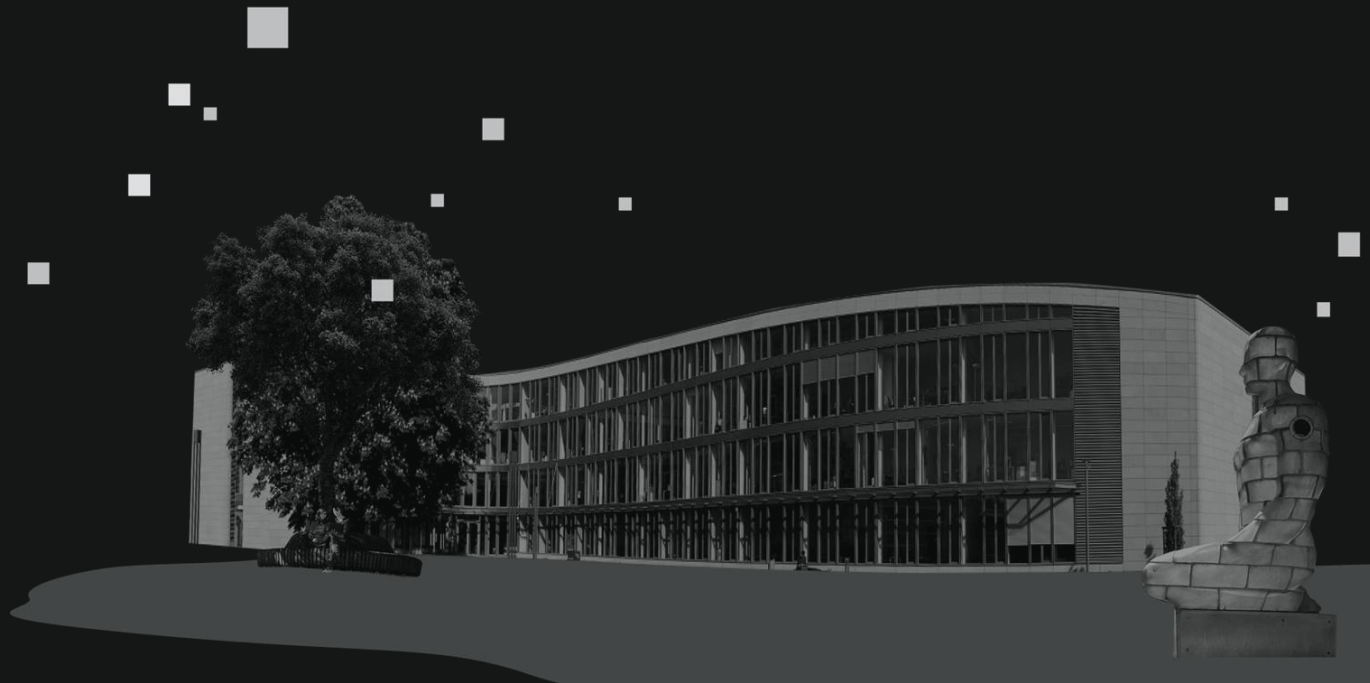
Hasso Plattner Institut
Digital Engineering · Universität Potsdam

Default Credentials Scanning

LLM-supported application-level default credential scanner

Laurenz Pepe Simon & Mathis Zscheischler

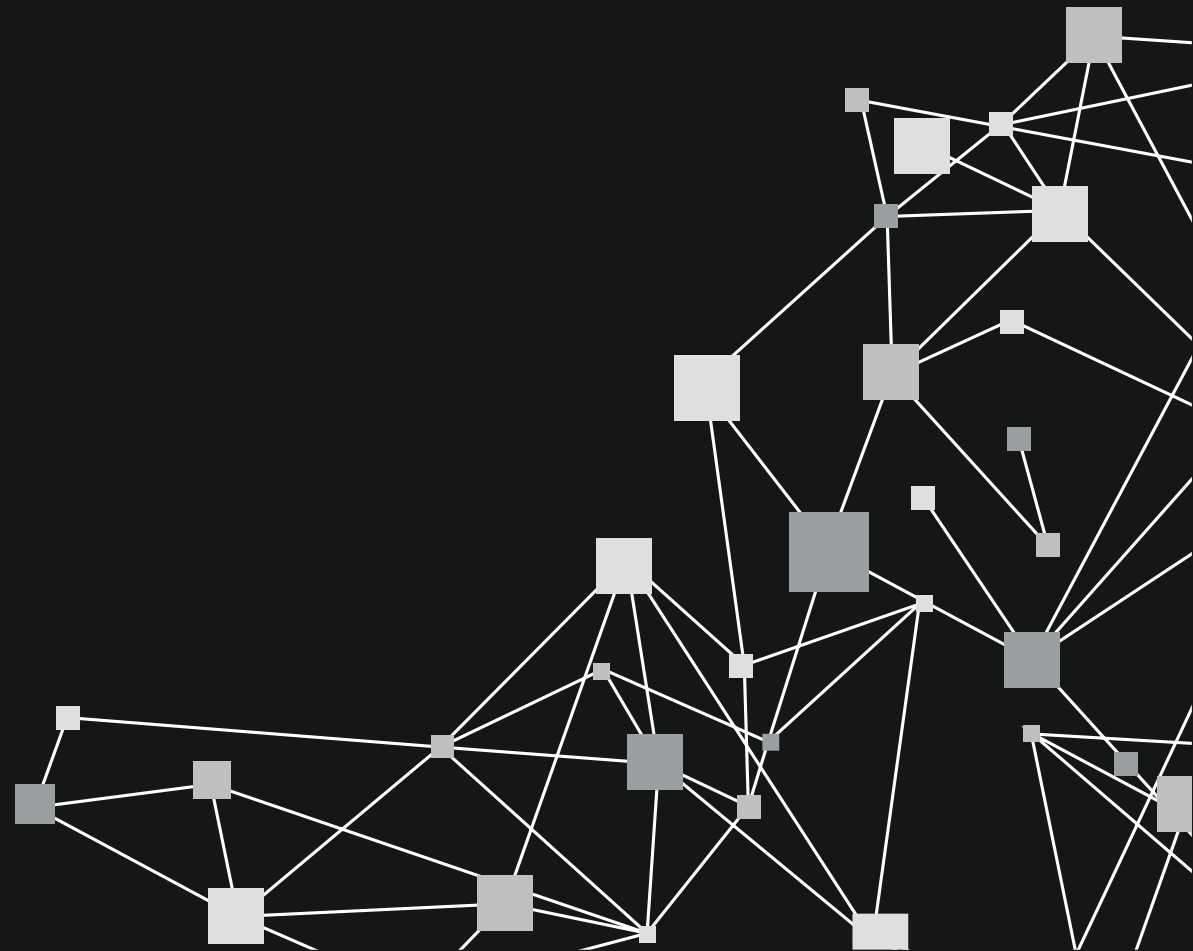


Design IT.
Create Knowledge.

www.hpi.de

HPI Hasso Plattner Institut
Digital Engineering · Universität Potsdam

Motivation

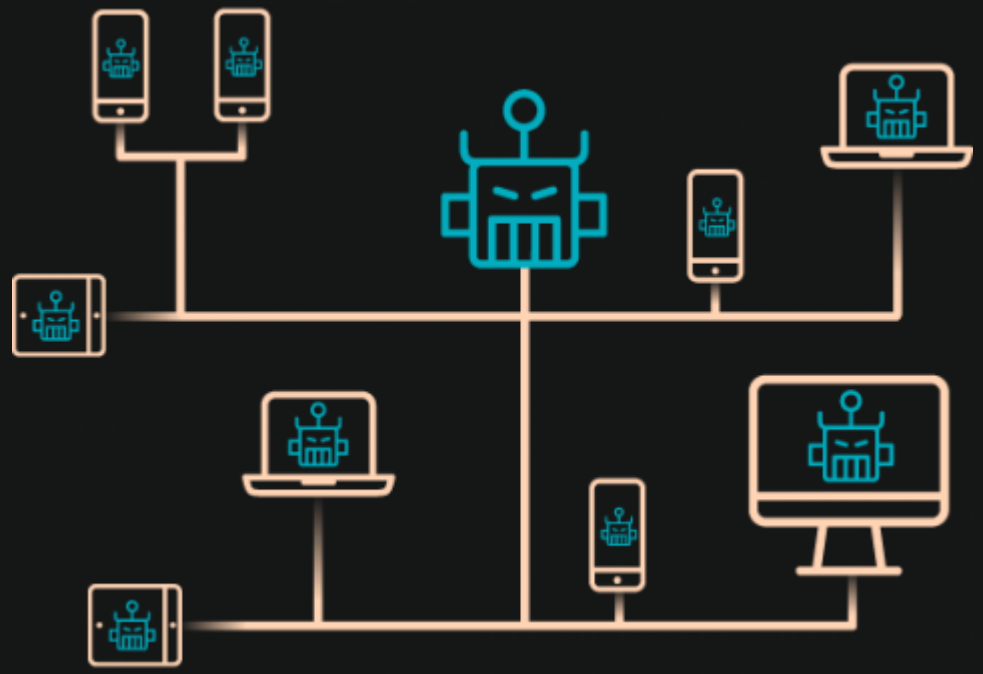


Design IT.
Create Knowledge.

www.hpi.de

Mirai Botnet — Massive DDoS Attacks (2016)

- Scanned for IoT devices with default credentials
  - Added each to botnet
- botnet grew to hundreds of thousands of devices
  - record-setting DDoS attacks
  - outages for Twitter, Netflix, Reddit…



https://www.myrasecurity.com/de/knowledge-hub/mirai/

McDonald’s AI Chatbot Default Password Exposure (2025)

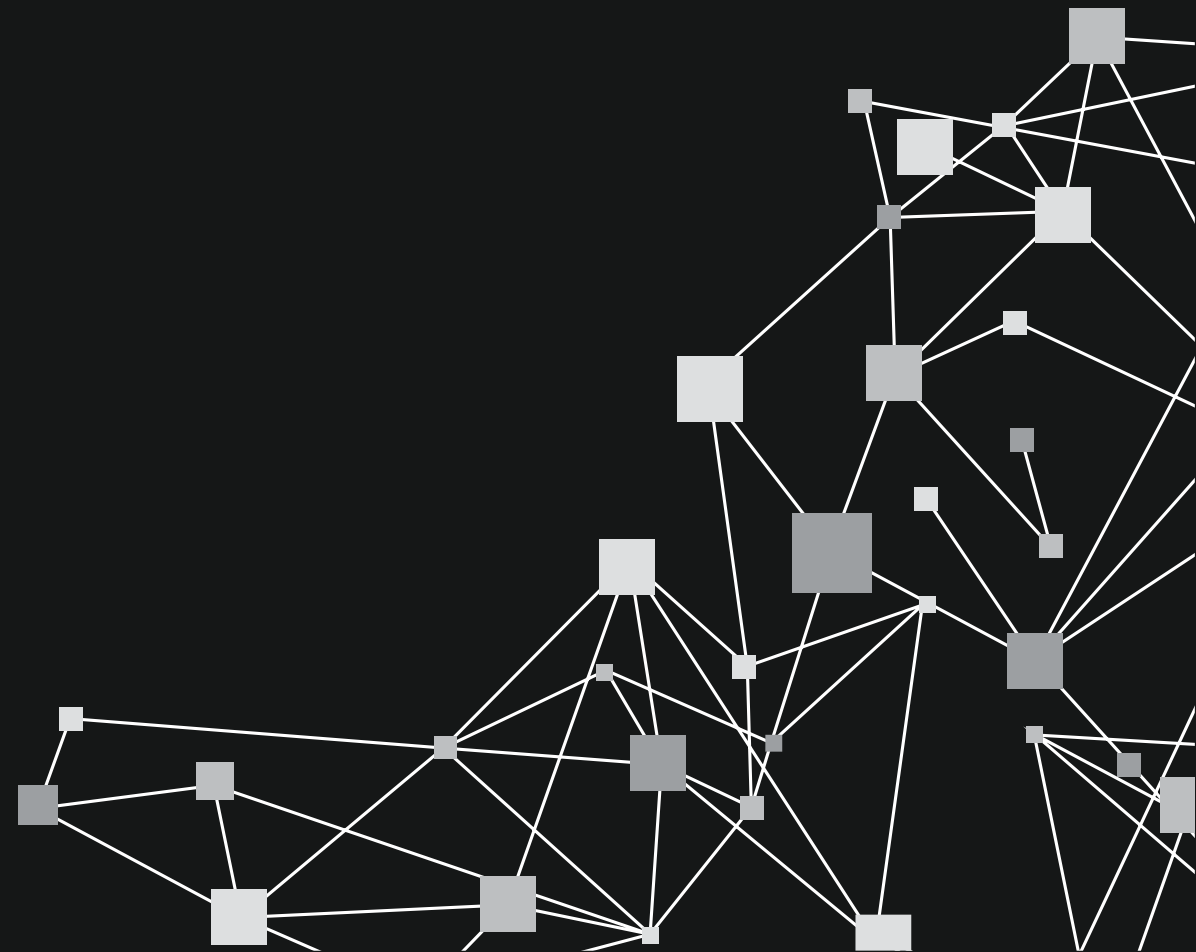
- third-party hiring chatbot used by McDonald’s retained a default admin password
  - access the system and view personal data of millions of applicants
  - exposing 64 million+ profiles

“the McHire administration interface for restaurant owners accepted the default credentials 123456:123456”

https://www.news.com.au/technology/online/hacking/dystopian-mcdonalds-ai-chat-bot-olivia-hacked-with-simple-password/news-story/2c0f09f9fb0396f6ca17c56419133fee

Hasso Plattner Institut
Digital Engineering · Universität Potsdam

State of the Art



Design IT.
Create Knowledge.

www.hpi.de

Default Credential Lists

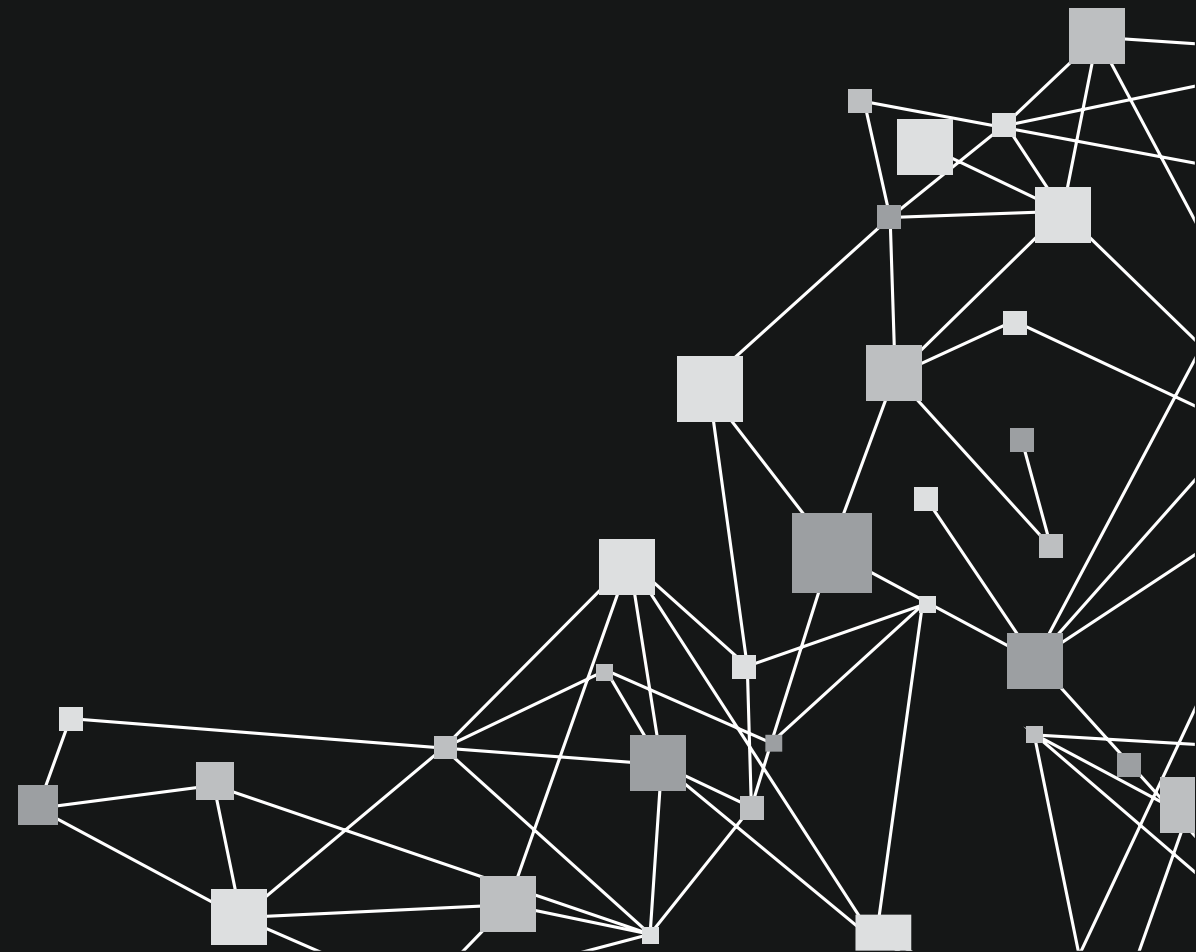
- https://default-password.info/
- https://github.com/danielmiessler/SecLists
- https://www.routerpasswords.com/
- ...

Apps / Tools

- Nmap http-default-accounts script
- Metasploit
- Vulnerability scanners (Nessus, OpenVAS, etc.)
- All rely on manually maintained lists

HPI Hasso Plattner Institut
Digital Engineering · Universität Potsdam

Project



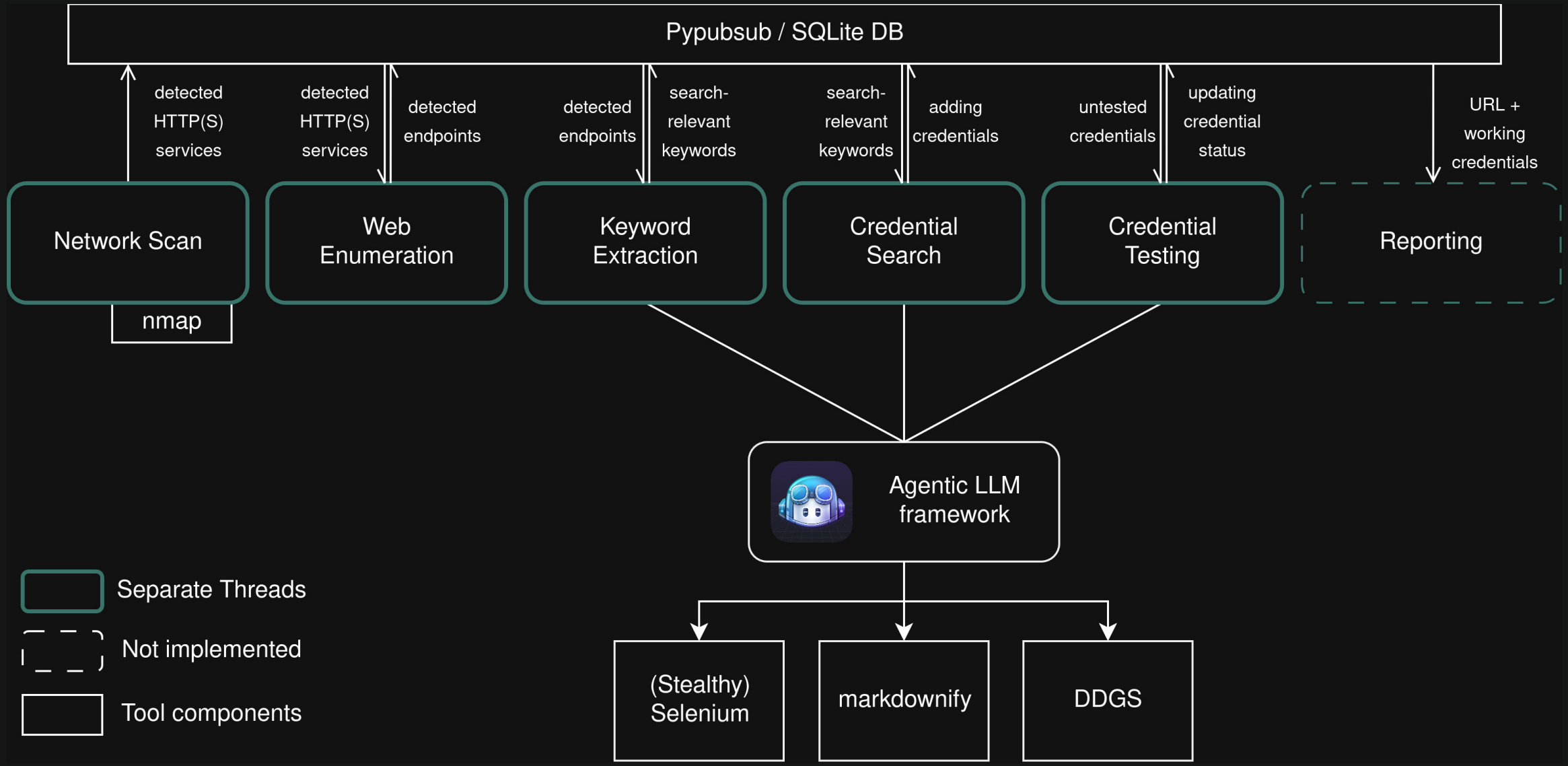
Design IT.
Create Knowledge.

www.hpi.de

Main Idea

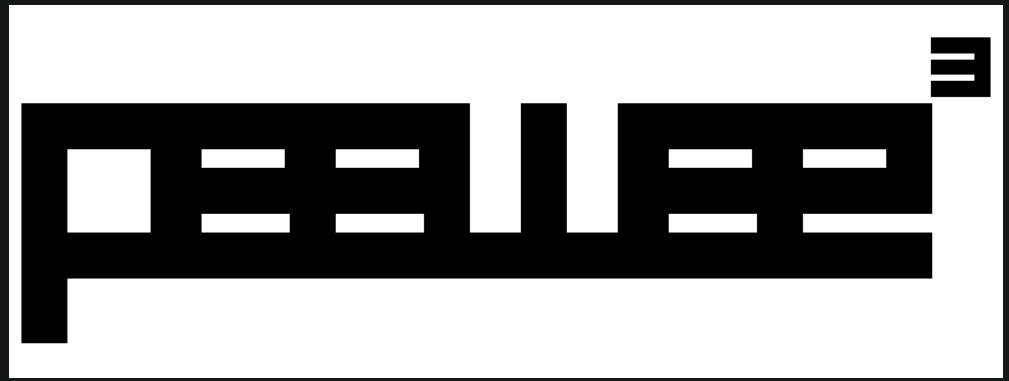
- Cover diverse applications using LLMs
  - Find credentials using agentic online search
  - Test credentials using agentic browser automation
- Build long-running scanner
  - Stateful
  - Runs continuously in background

Architecture



Architecture – Database

- Keep persistence and cache data
  - Want to persist over restarts/crashes
  - LLM calls are expensive (time & money)
  - Avoiding unnecessary network traffic
- Solution: SQLite database
  - Python package “peewee” allows easy database integration
- Structures communication between models



Architecture – Database

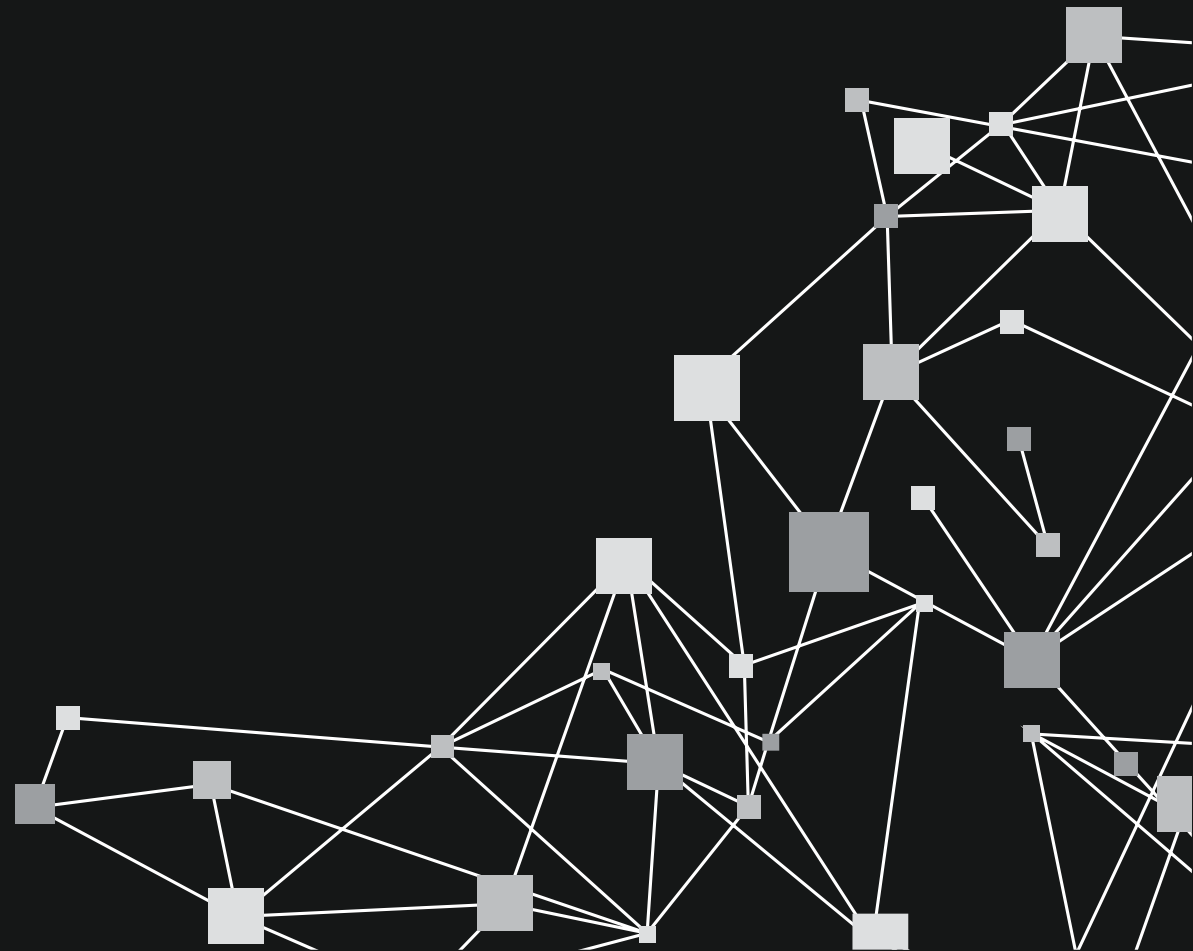
```python
class Service(BaseModel):
    """A discovered service with host, port and https true/false"""
    pk = pw.AutoField()
    host = pw.CharField(max_length=64)
    port = pw.IntegerField()
    https = pw.BooleanField()
    enum_in_progress = pw.BooleanField(default=False)
    _credentials = pw.TextField(null=True, default=None) # None means not searched yet
```

```python
class Endpoint(BaseModel):
    """A path on a service that responded with a non-error status code"""
    pk = pw.AutoField()
    service = pw.ForeignKeyField(Service, backref='endpoints')
    path = pw.CharField()
    is_login = pw.BooleanField(default=False)
    page_source = pw.TextField() # raw HTML of the page at this endpoint
    # keywords from the page at this endpoint; None means not analyzed yet
    _keywords = pw.TextField(null=True, default=None)
    _tested_credentials = pw.TextField(default="[]") # for login panels only
    working_credentials = pw.CharField(default="")
```

Architecture – Future Modules

- Reporting module
  - Informs admin about successful logins
- Subdomain enumeration
  - Some webapps only accessible with correct (sub)domain
  - There are tools for that
    - Not implemented yet

Difficulties (so far)



Difficulties – Finding Login Panels

- Invalid / Unreliable status codes
  - Server might always respond 200
  - Solution: query known invalid and compare with fuzzy-matching
- Client-side rendering
  - Single Page Applications might build DOM dynamically
  - Solution (TODO): Use Selenium instead of simple http requests

Difficulties – Finding Login Panels

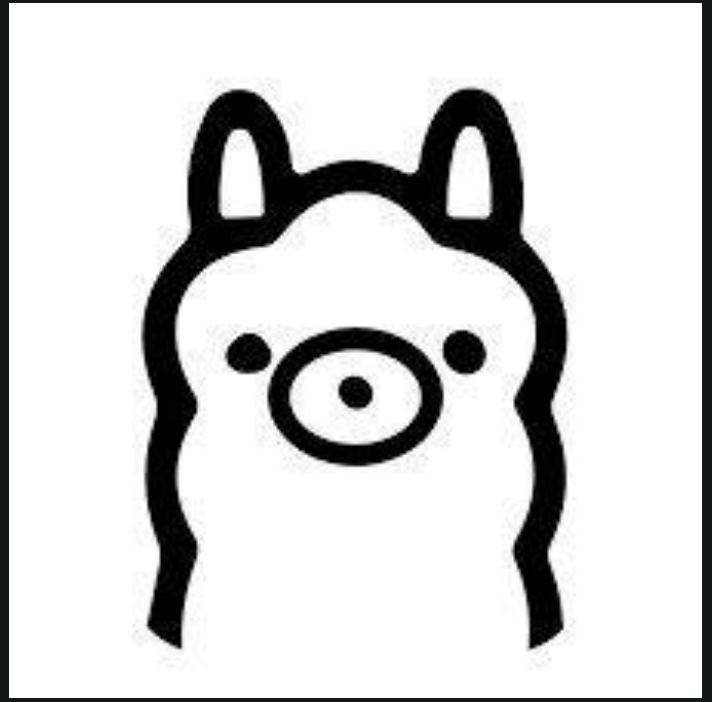
- Returning same page for many routes
  - Server might always respond with login panel if unauthenticated
  - Solution (TODO): fuzzy deduplication
- Subdomains
  - Multiple servers can share one port via reverse proxy
  - Solution (TODO): Fuzz "Host" header

Difficulties - AI

- With remote API
  - rate limiting
    - Parallel accesses can trigger "secondary rate limiting"
    - Solution: serialize API calls and respect rate limiting headers
  - sharing sensitive information
  - can be expensive

Difficulties - AI

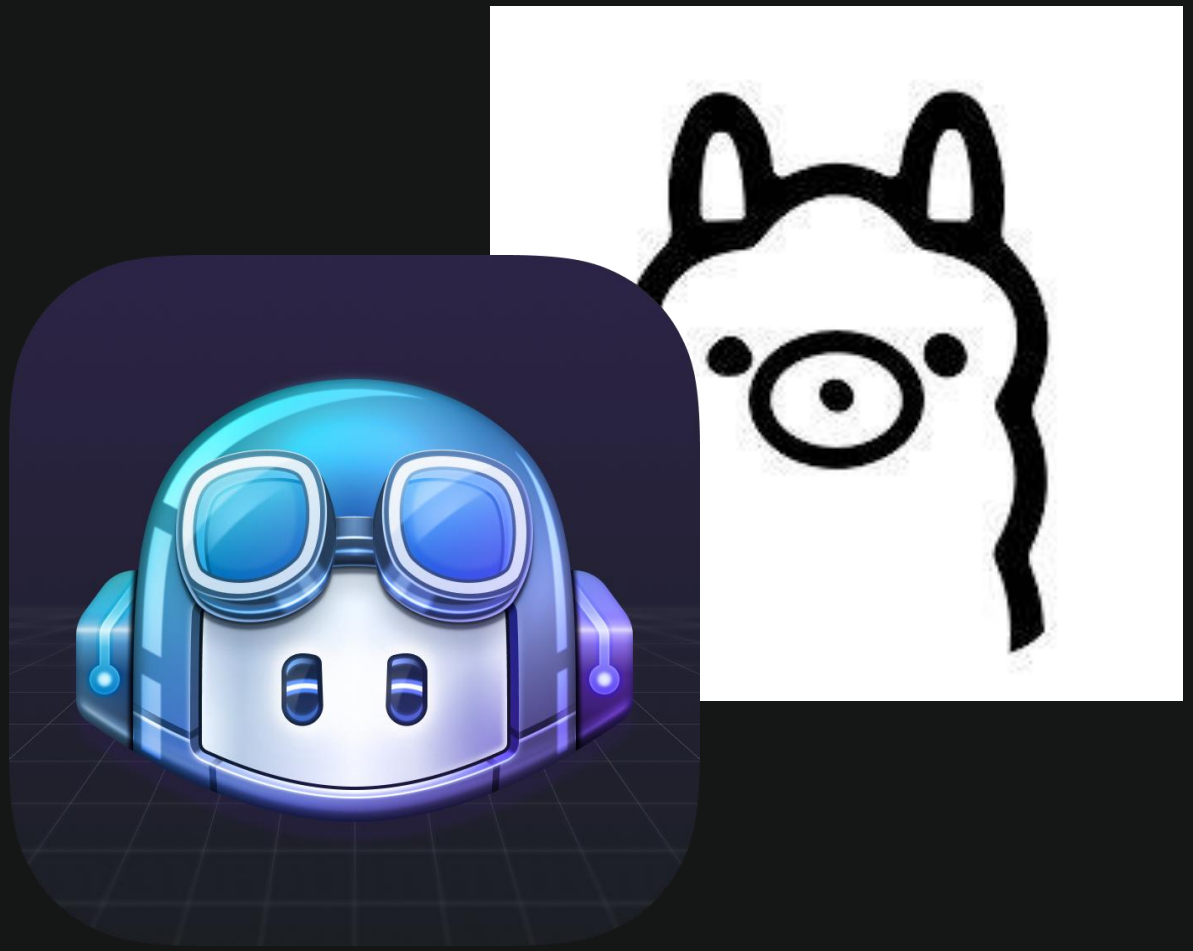
- With self-hosting (e.g., Ollama)
  - Very slow
    - GPU/NPU might not be available
  - Limited (V)RAM
    - Should assume conservative amount



Difficulties - AI

- Exceeding context window (CW) or max input tokens
  - Small LLMs limited to 32k – 128k CW
  - Github API limits to only 8k!

  => Context engineering required



Context Engineering

- Websites often large
  - Convert to markdown
  - Remove unnecessary information (e.g., scripts)
  - Depending on goal: summarization in chunks

Difficulties – LLM gets lost / forgets the task

- Good prompts are key
  - Bad: “Find default credentials for mailcow”
  - Can result in “I don’t know” or “I am not allowed to tell you that”
  - The LLM gives up quickly or isn’t using the provided tools

Difficulties – LLM gets lost / forgets the task

- Our current credential search prompt:

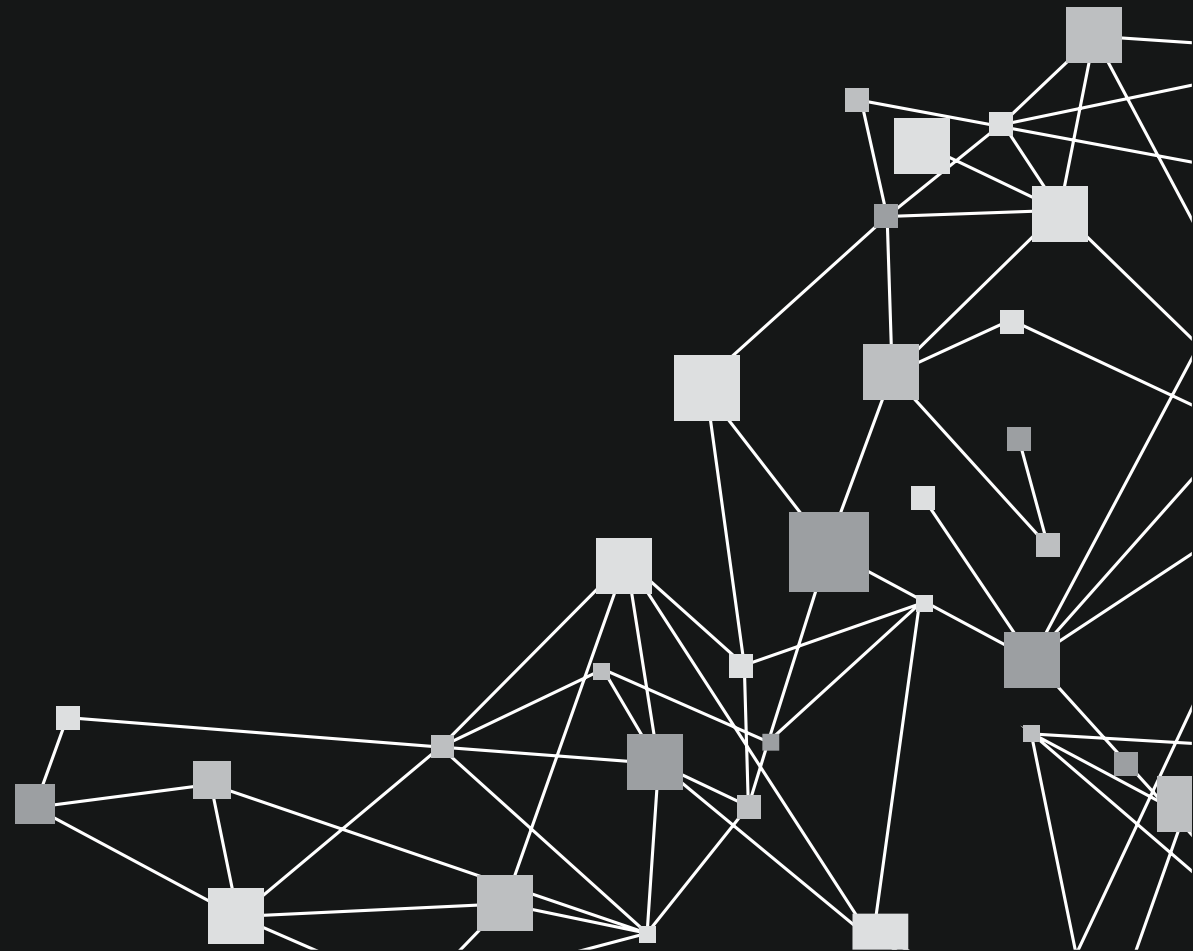
```
PROMPT_TEMPLATE = """
You are a blueteam pentester trying to find default credentials for an unknown web application.
From the web applications pages you have gathered the following keywords/links for a web search:
%s

Follow these steps to find the credentials:
1. Identify the official website or repository for the application.
2. Crawl the site or repository for any documentation, installation guides, or configuration files that might contain
default credentials. Follow links if necessary.
3. Submit the found credentials using the submit_credentials tool.

Search the web using carefully chosen, descriptive queries.
Use the fetch_url tool to retrieve the content of promising links.
If you find unrelated content, try refining your search.
You may use tools multiple times. Do not give up quickly.
Submit the credentials using the submit_credentials tool once you find them.
"""
```

Difficulties – Detecting Successful Login

- Heuristic to determine if login was successful
  - Redirection changes path
  - Content changes significantly
- False Positives!
  - Solution: record baseline using “known” wrong password + Cookies

```
login_successful = (
  (before_path != after_path) or
  (simhash.Simhash(before_page_source).distance(simhash.Simhash(after_page_source)) > 10)
)
```

Live Demo

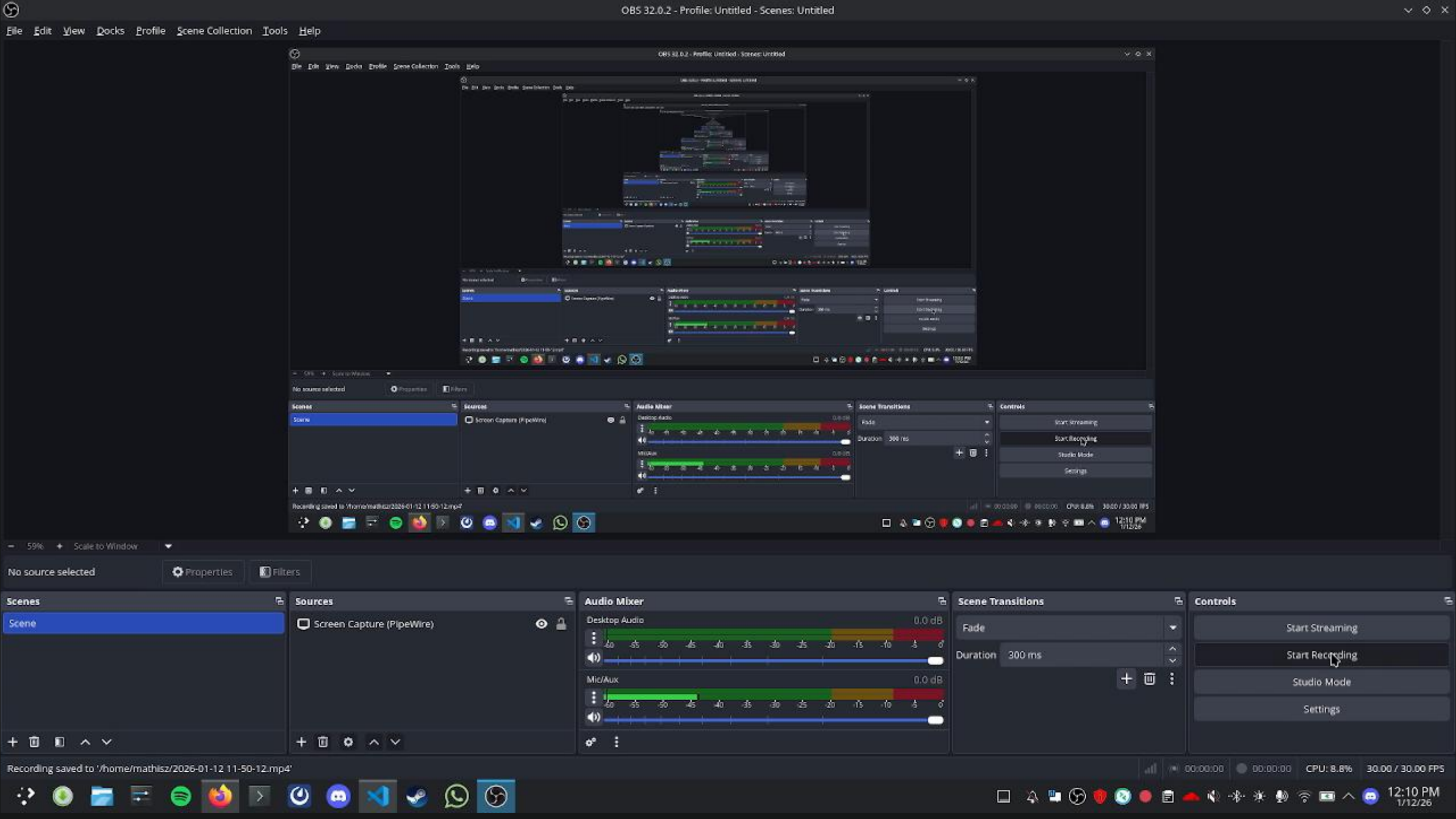


Design IT.
Create Knowledge.

www.hpi.de

MailCow

- Default credentials: admin:mohoo
- https://localhost:80


OBS 32.0.2 - Profile: Untitled - Scenes: Untitled
File Edit View Docks Profile Scene Collection Tools Help
No source selected
Properties
Filters
Scenes
Scene
Sources
Screen Capture (PipeWire)
Audio Mixer
Desktop Audio 0.0 dB
Mic/Aux 0.0 dB
Scene Transitions
Fade
Duration 300 ms
Controls
Start Streaming
Start Recording
Studio Mode
Settings
Recording saved to '/home/mathisz/2026-01-12 11-50-12.mp4'
CPU: 8.8%
30.00 / 30.00 FPS
12:10 PM
1/12/26

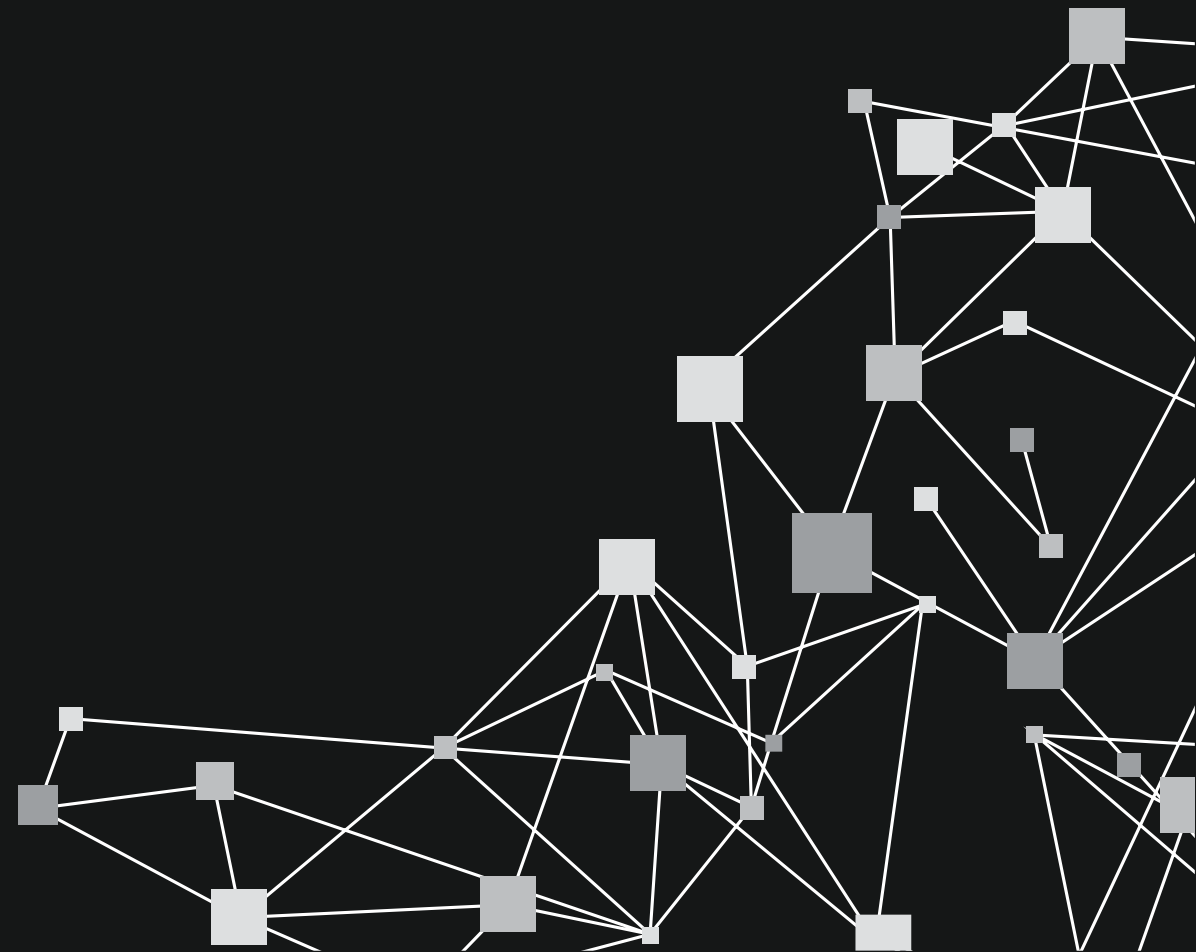
Outlook

Week	Tasks
19.01.-25.01.	Fully automated testing via CI/CD + 20 apps with default creds
26.01.-01.02.	Try and evaluate other methods/prompts using the testing setup
02.02.-08.02.	Implement persistence/long time scanning
09.02.-15.02.	Evaluation + Buffer
16.02.-22.02.	Final Presentation

TESTING	FUNCTIONAL	DEADLINE

Q&A

Do you agree with our concept, or would you approach it differently?



Design IT.
Create Knowledge.

www.hpi.de